

SafeLink: AI-Powered Mobile Platform Enhancing Public Safety and Community Policing in Sri Lanka

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Abstract—In Sri Lanka, community policing and public safety are dealing with more difficulties. These include slow emergency responses, road accidents, gender-based violence, and the rapid spread of harmful content on social media. Current safety and monitoring methods are mostly manual, reactive, incompatible, and limiting real-time risk detection and proactive intervention. This limits the ability to detect risks in real time and to take proactive measures. This study proposes an Artificial Intelligence (AI)-powered mobile platform to improve public safety by integrating various intelligent subsystems into a single framework. The platform includes four main features: an AI-based module for predicting risk to women and children, an emergency SOS feature with real-time location tracking and live streaming, an AI system to detect harmful social media content, and accident prediction using computer vision and machine learning techniques. By integrating computer vision, natural language processing, deep learning, and predictive analytics into a mobile-friendly design, the suggested system can help identify risks early, send timely alerts, and support data-driven decisions. This solution aims to enhance emergency response, assist in preventive policing, and promote safer designs for the Sri Lankan environment.

Keywords— *Public Safety, Community Policing, Emergency SOS, Traffic Violation Detection, Risk Prediction*

I. INTRODUCTION

Public safety is now a major issue in many developing countries due to rapid urban growth, rising vehicle density, expanding populations, and emerging social media and technological challenges. In Sri Lanka, problems like road traffic accidents, risk to women and children, slow emergency response, and the fast spread of harmful content on social media still threaten community well-being [1]. While law enforcement agencies and emergency services exist, many current systems are reactive, disjointed and rely heavily on manual processes. This limits their effectiveness in detecting and preventing risk in real time [2].

Traditional public safety and policing methods in Sri Lanka mainly respond to incidents. They rely on manual monitoring, citizen complaints, and investigations after events happened. Traffic enforcement depends on police presence and has limited Closed-Circuit Television (CCTV) coverage. Emergency response systems often lack real time situational awareness. Online threats are typically handled only after they

turn into real-world problems. These fragmented systems weaken prevention efforts and slow down timely responses, especially in high risk and quickly changing situations. Vulnerable groups, particularly women and children, face greater challenges because of lack of early risk detection and prevention safety measures [3],[4].

Recent progress in artificial intelligence, mobile computing and data analytics creates new chances to improve public safety systems through real time monitoring, predictive modeling and smart automation. AI-based methods have delivered promising results in several areas, including traffic violation detection using computer vision, accident prediction with machine learning, natural language processing for detection harmful content on social media and mobile emergency response systems. However, most existing solutions target specific problem areas in physical-environment monitoring. The lack of integration among physical-world monitoring, digital threat analysis, emergency response, and community involvement limits their overall effectiveness, especially in resource-limited environments like Sri Lanka.

The first feature is an AI-powered system for predicting safety risks for women and children. It uses contextual, temporal, and behavioral data to spot high-risk situations and encourage early preventive action. Previous studies show that predictive analytics and risk modeling can significantly improve identifying unsafe environments and potential threats to vulnerable groups. The proposed system brings together different data analysis methods by combining video-based object detection, speech-to-text processing, and multilingual natural language classification to improve the accuracy and reliability of identifying incidents. This supports proactive safety forecasting and data-driven preventive measures for women and children [5].

Emergency response systems in Sri Lanka, including national police hotlines, mainly rely on voice communication. In high stress or life-threatening situations, victims often have difficulty clearly stating their location, the nature of the incident or the risk level. Most current SOS systems and mobile safety apps react to situations. They offer little predictive insight and lack sufficient awareness of risk factors [6]. Furthermore, the effectiveness and accessibility of such

systems in the Sri Lankan context are diminished by language barriers and a dearth of locally relevant data-driven insights [7]. The significance of proactive safety prediction models that can identify high-risk circumstances prior to incidents happening is highlighted by these difficulties.

Another growing challenge is the quick spread of harmful and crime-related content on social media platforms. Incidents of abuse, violence and threats are often first reported online instead of through formal law enforcement channels. This leads to delays in intervention [8]. Manually monitoring social media content is impractical because of the huge volume and speed of data generation. Additionally, automated systems that rely on a single type of input often miss the contextual complexity of harmful behavior [9]. Artificializes show that combining text, audio and visual analysis with multimodal artificial intelligence can significantly improve detection accuracy and response speed [10].

Road traffic accidents are one of the top causes of injury and death in Sri Lanka. They lead to thousands of deaths and large economic losses every year. AI-based traffic monitoring systems that use object detection and license plate recognition have yielded promising results in automating enforcement and improving road safety [11]. Likewise, machine learning techniques have been effectively applied to analyze crime hotspots, prioritize emergency risks and forecast safety [12]. However, most existing solutions handle these challenges separately and do not interfere with a unified, mobile-friendly platform designed for local social, cultural and infrastructure conditions.

To address these gaps, this research suggests an AI-powered mobile platform. The goal is to improve public safety and strengthen community policing in Sri Lanka by integrating several smart subsystems. The platform includes four main features: predicting safety risks for women and children, an emergency SOS feature with real time location tracking and live streaming, detecting harmful content on social media in real time and smart traffic violation detection with accident predictions. By bringing these features together in one mobile framework, the system seeks to enable proactive risk identification, timely alerts and data driven decision making for both citizens and policies. This unified approach helps create scalable, context aware public safety solutions that fit Sri Lanka's social, linguistic and infrastructure needs.

II. LITERATURE REVIEW

A. Public safety and the Sri Lankan context

Public safety in Sri Lanka mostly relies on reactive measures like emergency hotlines and post-incident reporting systems. While these services offer immediate response help, they do not have predictive intelligence and data-driven forecasting abilities. Recent research shows that predictive analytics and machine learning models can spot high-risk areas by examining past and contextual crime data. This allows for proactive intervention instead of a delayed response [13].

Another important challenge in Sri Lanka is language diversity. Incident reports and online content often appear in a Sinhala-English mix, which reduces the effectiveness of systems that use only one language. Research on multilingual and code-mixed natural language processing shows that localized AI models are needed for accurate classification of safety-related content. Additionally, approaches that combine

video, audio, and text analysis have shown better detection reliability than systems using just one type of data, especially in safety-critical environments [14], [15].

Although global studies confirm the effectiveness of predictive and multimodal safety systems, there is still a gap in creating an integrated, bilingual, and context-aware framework specifically for Sri Lanka. The proposed research aims to fill this gap by using predictive modeling, multilingual Natural Language Processing (NLP), and multimodal AI techniques to improve proactive protection for women and children.

B. Women and children safety prediction and risk mapping

Traditional safety systems are largely reactive (e.g., SOS after an event), while recent research trends move toward predictive safety using machine learning, geospatial analysis, and spatio-temporal modeling. Crime hotspot prediction studies demonstrate that clustering-based hotspot identification and time-aware features can support forecasting and targeted interventions [16]. Broader reviews of spatio-temporal hotspot detection highlight the value of combining spatial clustering with temporal trend modeling to improve predictive performance and interpretability for decision support [17].

For Sri Lanka, multilingual and code-mixed reporting is a practical requirement because users often mix Sinhala and English in informal reports. Code-mixed NLP research has shown the feasibility of Sinhala-English modeling, including handling code-mixed inputs for downstream language tasks (e.g., translation pipelines that normalize code-mixed content) [18]. This supports bilingual risk-report understanding and classification components that aim to improve accessibility and inclusiveness.

C. Emergency SOS with real-time location and live streaming

Emergency response in Sri Lanka mainly depends on hotline services like the 119-emergency number. These services rely heavily on voice communication between victims and responders. Prior studies show that voice only emergency reporting often fails in high stress situations because of panic, injury, language barriers or fear. This leads to delayed responses and incorrect information delivery. Survey findings from this study further confirm these issues, showing that many people struggle to explain emergencies and provide exact location details. Although early mobile SOS applications introduced Global Positioning System (GPS)-based location sharing to improve response efficiency, these systems offer only static location data and lack contextual awareness. This limits responders' ability to understand the severity of incidents and priorities resources.

To address these issues, recent research has investigated multimodal emergency communication systems that combine real time video and audio streaming with location data. International efforts like Europe's Next Generation 112 (NG112) and Australia's BluLink show that sharing live multimedia greatly improves situational awareness and emergency response effectiveness [19]. However, many of these systems rely on manual monitoring by human operators, which restricts scalability during multiple emergencies. Advances in AI help fill this gap with automated analysis. Computer vision models like YOLOv11 enable real time detection of weapons, fire and smoke [20]. Meanwhile, audio models like YAMNet recognize distress

related sounds such as shouting and alarms [21]. Multimodal analysis has been proven to improve emergency prioritization and reliability when compared to single modality methods.

D. Harmful social media content detection and multimodal moderation

Recent studies have shown that advanced language models, such as transformers, are very good at detecting harmful content on social media. Their strength lies in understanding context, which helps them catch abusive or violent language that might otherwise go unnoticed. Multilingual models such as XLM-RoBERT are especially useful in places like Sri Lanka, where people often mix Sinhala and English in their posts. Unlike older machine learning tools, these models can understand the meaning behind words and phrases, making them much better at identifying different types of harmful language.

In addition to checking text, researchers are now using computer vision to spot signs of violence in images and videos. Real-time systems like YOLO can quickly detect weapons and other dangerous objects in photos and video streams. When you combine tools like YOLO with language models, you get a much more accurate way to flag harmful content. This helps social media platforms step in earlier and keep their communities safer.

E. Smart traffic violation detection and accident-risk analytics

AI-based intelligent transportation systems commonly rely on computer vision for vehicle detection, violation identification, and enforcement support. Modern object detection methods such as YOLOv4 are widely used due to YOLO's balance of speed and accuracy, making them suitable for near real-time detection in surveillance-like settings. In addition, license plate recognition (LPR) remains a key enabling technology for enforcement workflows, and survey literature outlines standard LPR pipelines and common challenges under varying illumination, motion blur, and viewing angles.

Accident risk prediction and enforcement analytics frequently incorporate spatial patterns and temporal features, echoing findings from hotspot prediction research where historical events and time-of-day effects influence risk. Sri Lanka's reported patterns in transport injuries also highlight that accidents vary by day and time, motivating time-aware predictive modeling. These works collectively support traffic modules that integrate detection (violations) and prediction (risk-prone zones) to enable prevention, not only post-event analysis.

III. METHODOLOGY

The overall system consists of four main features: Women and children safety, Emergency SOS with real-time location and live streaming, Harmful social media content detection and multimodal moderation feature, Smart traffic violation detection and accident prediction system. Each component is organized in the system as shown in the following system diagram.

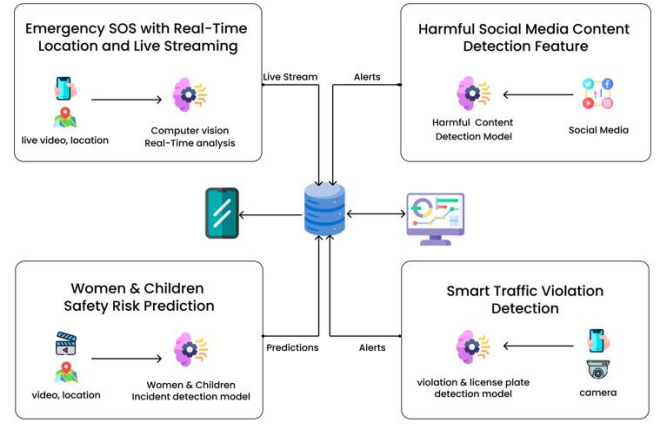


Fig. 1. Overall system diagram

A. Women and children safety prediction and risk mapping

This research proposes a multimodal AI methodology to analyze incident data related to women and children and to predict potential safety risks in advance. The system changes raw multimedia incident data into structured safety information by combining video analysis, audio speech processing and bilingual text classification. Incident data comes from a centralized reference database that contains historical emergency and public safety records, including video recordings, extracted audio segments, and textual descriptions. Preprocessing steps such as video normalization, audio cleaning and Sinhala-English text standardization are used to ensure data consistency and reliability.

The analysis starts by identifying incidents that involve women and children. We process video data with a YOLO-based object detection model that is trained to spot women, children, and important context. We convert audio from the video files into text using a speech recognition model, which helps us identify distress-related language. Then, we analyze the transcripts and text reports with a NLP model based on XLM-RoBERTa. This helps us classify incidents into categories like harassment, unsafe public transport, child endangerment, or normal activity.

The A multimodal fusion mechanism combines outputs from video, audio, and text modules to improve reliability. It integrates confidence scores and contextual features from each modality. The structured results are stored in a database with metadata such as time, location, and incident type. Analysts then review these stored incidents to find patterns related to risk in space and time. This helps estimate high-risk areas and times of peak vulnerability. Based on these patterns, the system provides safety insights, including risk scores, heatmaps, and proactive alerts sent through a mobile app and police dashboard. This approach shifts the safety framework from reactive reporting to proactive risk identification for women and children.

B. Emergency SOS with real-time location and live streaming

The proposed Emergency SOS system uses a multimodal, AI-assisted approach to improve real time emergency response. The setup combines one touch SOS activation, continuous GPS tracking, live video streaming, automated threat detection and risk based prioritization within a mobile backend.

SOS activation, the mobile application automatically captures the user's real time GPS coordinates and initiates live video streaming. All data are time-stamped and geo-tagged to preserve contextual integrity. The multimedia streams are securely transmitted to a backend server, where alerts are validated before processing.

Video frames are analyzed with the YOLOv11 object detection model to spot threats like weapons, fire, snake and visible injuries. Audio streams are processed through the YAMNet model to pick up distress sounds such as shouting, crying, and alarms. Detection outputs produce confidence scores that are combined with historical crime data to calculate a severity based risk score. Validated alerts, live streams, and risk levels appear on a secure police dashboard, while encrypted geo-tagged evidence is kept for accountability and future investigation.

C. Harmful Social media content detection and multimodal moderation

This research introduces a system that can spot harmful content on social media in real time, helping make communities in Sri Lanka safer and allowing for quick action when needed. The system automatically looks for things like harassment, hate speech, abuse against children and older adults, violent acts, and even threats of murder across text, audio, and video posts.

The system collects data directly from social media feeds and analyzes it almost instantly. For text, it uses a smart language model that can handle both Sinhala and English, even when they're mixed. This model checks for different types of harmful language and assigns each a confidence score. For images and videos, another tool spots weapons, blood, or violent actions, and it checks frames over time to be more accurate. If there's audio, it turns spoken words into text so they can be checked in the same way. All results are combined into an overall risk score that appears on a live dashboard, so people can quickly review and respond to threats.

D. Smart Traffic Violation Detection and Accident Prediction

To improve automated traffic monitoring and road safety in Sri Lanka, this study proposes a real-time AI-based smart traffic violation detection and accident prediction system. Using computer vision techniques, the system analyzes traffic images and video streams and uses them to identify vehicle categories, detect violations, and recognize license plates.

CCTV cameras and user-uploaded media are used to collect traffic data. To increase robustness under different lighting and environmental conditions, preprocessing techniques such as resizing, normalization, and enhancement are used. Real-time vehicle and license plate recognition is performed using an object recognition model based on YOLOv11 that produces class labels with confidence scores. Non-maximal suppression is applied to remove redundant detections.

To extract alphanumeric characters from Sri Lankan license plates, the detected license plate regions are cropped and processed using PaddleOCR. Precision, recall, F1-score and mean average precision (mAP) are used to evaluate the performance of the model. For proactive traffic control, the

system allows for data-driven accident prediction and automatic violation monitoring.

IV. RESULT AND DISCUSSION

This section discusses how each feature and its deep learning models performed. We didn't just rely on training and testing accuracy; we also evaluated real-world accuracy in various lighting conditions, resource usage, and deployment potential.

A. Women and children safety prediction and risk mapping

In The proposed multimodal safety analysis system was evaluated using separately trained video and text models to assess its effectiveness in detecting women- and children-related incidents. The evaluation focused on model accuracy, precision-recall balance, and overall reliability under different confidence thresholds.

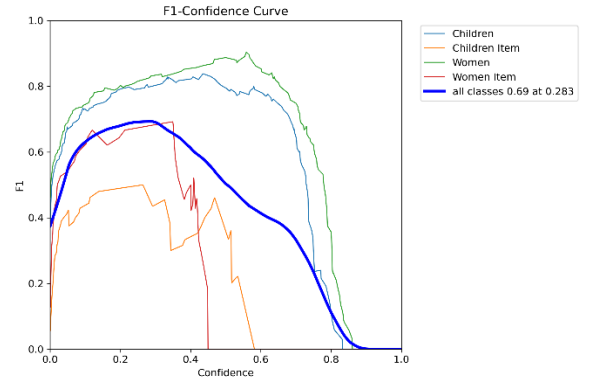


Fig. 2. F1 confidence curve computer vision model

For the video analysis part, we trained a YOLO-based object detection model on a custom-labeled dataset that included women, children, and related classes. As shown in Figure 2, after 60 training epochs, the model reached a testing accuracy of about 91.8%. Performance metrics such as precision, recall, F1-score, and mean Average Precision (mAP@0.5 $\approx$ 0.73) show that the detection performance is reliable. The precision-recall and F1-confidence curves reveal a stable balance between detection accuracy and completeness. This confirms that the model is suitable for real-time safety applications.

For text analysis, a multilingual XLM-RoBERTa model was trained on a Sinhala-English code-mixed dataset for 30 epochs. It achieved an initial testing accuracy of about 71.8%. Although this is lower than the video model's performance, it shows the difficulties posed by limited localized data and informal language patterns. The model offers a starting point for further improvement through dataset expansion and fine-tuning.

Audio processing with Whisper allowed effective speech-to-text conversion. This made it possible to classify incidents even when visual information was lacking. Combining video, audio, and text analysis made the system more reliable than those using only one type of data. Though performance differs among the various types of data, the results show that the proposed system is practical. They also emphasize the need for better datasets and improved combination of data for future development.

Overall, the results validate the feasibility of the proposed system in identifying women-related and children-related

safety incidents and demonstrate its potential to support proactive safety prediction and informed decision-making.

B. Emergency SOS with real-time location and live streaming

The proposed Emergency SOS system was assessed for its real-time data transmission, automated threat detection, and risk-based prioritization. The combination of continuous GPS tracking and live audio or video streaming gave responders immediate awareness of the situation. This approach greatly lowered confusion compared to traditional voice-only emergency reporting. Real-time location updates made it easier to identify the exact position of incidents, especially in high-stress situations where talking is not dependable.

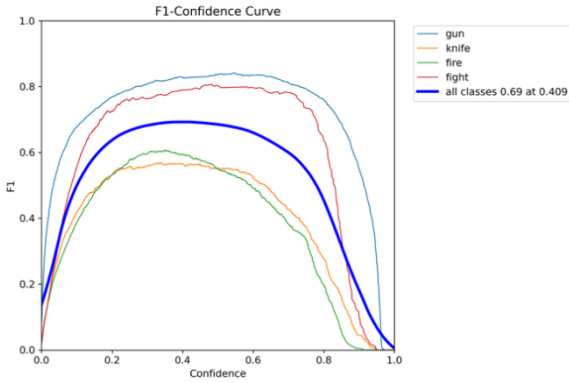


Fig. 3. F1 confidence curve computer vision model

As seen in Figure 3, the F1 Confidence Curve above, the YOLOv11 model effectively detects visual threats like guns, knives, fire, and fight situations. The graph indicates that the model achieves its highest F1 Score of about 0.69 at a confidence level of 0.409. This means the model performs optimally at this point by balancing correct detections with fewer false alarms. The performance for guns and fights is stronger than for knives and fire, likely due to the more distinct visual features in those categories. When the confidence threshold is set too high, the F1 score decreases because the model becomes overly strict and misses some detections.

In addition to video detection, the YAMNet model successfully detects distress sounds like shouting, alarms, and crying. Using both video and audio analysis improves the system's reliability. The risk scoring system then combines these detection results with historical crime data to prioritize serious incidents on the police dashboard. Overall, the system offers faster and clearer understanding of emergencies.

C. Harmful Social media content detection and multimodal moderation

The system for detecting harmful content on social media was tested on a dataset containing Sinhala, English, and mixed-language posts. It was designed to pick up on six types of harmful content: harassment, hate speech, child abuse, elder abuse, violence, and murder threats. Tests showed that the model could find harmful posts even when people used slang, misspellings, or a mix of languages, which is common on Sri Lankan social media.

The model performed well overall, correctly finding most harmful content. It was especially good at spotting direct threats and violent language, since these usually have clear warning signs. However, it sometimes struggled with more subtle forms of hate speech or sarcasm, which can be harder to detect because the language is less obvious or depends on cultural context.

In addition to analyzing text, the system uses computer vision to spot dangerous items, such as weapons, in photos and videos posted online. By combining information from text, images, and audio, the system makes more accurate decisions about what's truly harmful. This approach helps avoid mistakes that can happen if you only check the text.

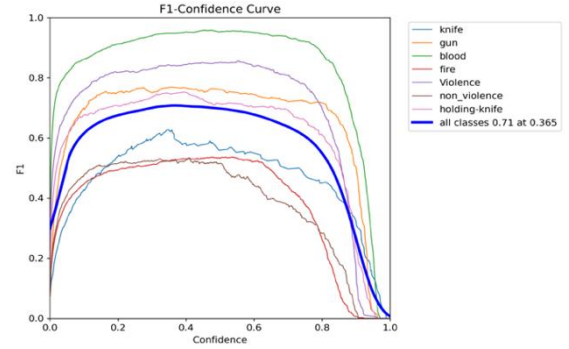


Fig. 4. F1 confidence curve computer vision model

Figure 4 illustrates the F1 Confidence Curve of the YOLO model across multiple classes, including knife, gun, blood, fire, violence, non-violence, and holding knife. The horizontal axis represents the confidence threshold used to accept detections, while the vertical axis shows the F1-score, which balances precision and recall. Each curve corresponds to a specific class, and the thicker blue line represents the overall performance across all classes. The graph indicates that the model achieves its best overall F1-score of approximately 0.71 at a confidence threshold of around 0.36. Classes such as blood and violence demonstrate stronger performance, while fire and non-violence show comparatively lower detection accuracy. As the confidence threshold increases toward 1.0, the F1-score decreases because the model becomes more selective and misses valid detections. This curve helps determine the optimal confidence threshold for balanced detection performance.

The results show that automatic tools like this can help monitor social media and step in quickly when needed. While the system works well, it could get even better with more local data and smarter ways to understand context and combine different types of information. Overall, this approach offers a strong way to make the internet safer and protect people online.

D. Smart Traffic Violation Detection and Accident Prediction

The proposed Smart Traffic Violation Detection and License Plate Recognition system was evaluated using a multi-class vehicle dataset containing front and back views of lorries, buses, cars, bikes, vans, and three-wheelers, along with number plate annotations. A YOLO-based object detection model was used for vehicle and plate localization.

For the license plate detection module, the model achieved a Precision of 0.9449, a recall of 0.9289, an

mAP@0.5 of 0.9714, and mAP@0.5–0.95 of 0.8347, indicating strong detection and localization performance. Figure 4 presents the confusion matrix, which shows clear diagonal dominance, confirming accurate classification across most vehicle categories. Minor misclassifications were observed between visually similar classes and background regions.

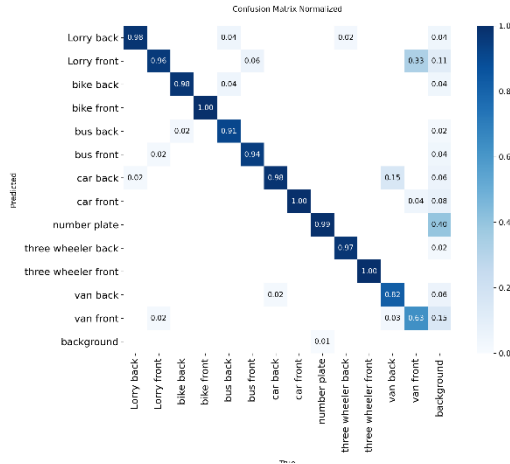


Fig. 5. Confusion matrix of the proposed vehicle and traffic violation detection model

For the traffic violation detection module, the model achieved a Precision of 0.8012, a recall of 0.7472, an mAP@0.5 of 0.8012, and mAP@0.5–0.95 of 0.5287. The slightly lower performance is mainly due to the contextual complexity of violation scenarios compared to direct object detection. The system successfully detects multiple vehicle types and number plates with confidence values generally between 0.8 and 1.0. Detected plate regions are processed using PaddleOCR, which accurately recognizes Sri Lankan vehicle registration numbers under varying lighting and background conditions.

Overall, the results demonstrate that the integrated YOLO detection and PaddleOCR recognition framework provides reliable vehicle identification and forms a practical foundation for automated traffic violation monitoring and accident risk assessment.

V. CONCLUSION

This paper presented an AI-powered mobile platform designed to enhance public safety and strengthen community policing in Sri Lanka through the integration of four intelligent subsystems: women and children safety risk prediction, emergency SOS with real-time location and live streaming, harmful social media content detection, and smart traffic violation detection with accident prediction. By combining computer vision, natural language processing, deep learning, and predictive analytics within a unified mobile-centric framework, the proposed system enables proactive risk identification and timely decision support.

Unlike traditional reactive approaches, the platform emphasizes real-time intelligence, multilingual support, and contextual awareness tailored to the Sri Lankan environment. The integrated architecture demonstrates the potential of AI-driven solutions to improve emergency responsiveness, support preventive policing, and promote safer communities in a scalable and practical manner.

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